Sujan Dora

AI Engineer

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Professional Summary

AI/ML Engineer with 2 years of experience building LLM-RAG based document QA and production-ready AI solutions. Delivered systems that improved document QA accuracy by ∼20%, reduced MFA verification load by ∼25%, and optimized ID authentication pipelines, with hands-on experience in Python, PyTorch, TensorFlow, OpenCV, and AWS. Recent graduate with strong foundation in LLMs and RAG systems, model evaluation, and end-to-end ML workflows.

Education

Missouri State University — GPA: 3.63
M.S. Computer Science – Data Science — Dec 2025

Technical Skills

Programming Languages	Python, C++
ML Frameworks & Libraries	PyTorch, TensorFlow
LLM & NLP Tools	Multimodal LLMs, Generative AI, RAG
Cloud & Deployment	AWS, Docker
Databases & Data Engineering	Data Generation, Attributes
Web & DevOps	CI/CD, Infrastructure
ML Concepts & Specializations	Perception, Scene Understanding, Model Validation

Work Experience

AI/ML Engineer — Chennai, India
Grootan Technologies — July 2023 - December 2023

- Implemented a self-correcting RAG system using LangChain, a foundational architecture for developing advanced intelligent systems like multimodal agents.
- Built a hybrid retrieval workflow (Dense + BM25), optimizing context retrieval which is critical for scenario understanding tasks.
- Developed an LLM-as-Judge evaluation pipeline to validate model outputs, ensuring reliability in determining complex environment states.
- Engineered batched query extraction pipelines to improve inference efficiency, relevant to processing large-scale sensor/driving data.
- Designed ML-based anomaly detection systems for verification, aligning with the need to identify hazards and driving restrictions.
- Deployed scalable AI models using Docker and AWS, ensuring robust infrastructure for offline driving intelligence workflows.

AI/ML Intern — Chennai, India
Grootan Technologies — September 2022 - July 2023

- Built a National ID authentication system for detecting counterfeit holograms using YOLO object detection, MiDaS depth estimation, and GAN-generated synthetic samples to improve robustness.
- Optimized the hologram verification pipeline by eliminating redundant processing, reducing per-card detection time from 7 minutes to 3 minutes.
- Applied transfer learning to fine-tune pre-trained models for limited datasets, achieving reliable counterfeit detection across varied hologram designs.
- Containerized and deployed solutions using Docker and AWS, ensuring scalable production-ready pipelines for anomaly detection, identity verification, and document QA systems.

Data Science Intern — Bengaluru, India
iNeuron.ai — November 2021 - March 2022

- Developed an end-to-end credit card default prediction project using Docker, CircleCI, and Heroku, achieving 87% model accuracy on simulated customer transactions.
- Built a CI/CD pipeline to deploy predictive models to AWS production, enabling real-time predictions.
- Conducted exploratory data analysis on 30,000+ customer records to identify key features that improved model performance.
- Implemented data validation and preprocessing pipelines, reducing manual data preparation effort.

Personal Projects

Wine Quality Prediction - End-to-End MLOps Pipeline

- Built an end-to-end ML pipeline using ZenML and MLflow, deploying a REST API on AWS Lambda with Docker for automated training and inference.
- Implemented CI/CD with GitHub Actions to automate model retraining and deployment to AWS (S3, ECR, API Gateway), ensuring reliable and repeatable releases.

BioMedical Research Assistant - Multi-Agent LLM System

- Fine-tuned Llama-3.1-8B using QLoRA on PubMedQA data with 4-bit quantization, gaining hands-on experience in parameter-efficient LLM training for biomedical QA.
- Built a multi-agent research workflow using CrewAI and PubMed retrieval via Model Context Protocol, with a Streamlit interface for interactive querying.

SafeSpace 2.0 - AI Mental Health Therapist

- Developed an AI-powered mental health companion with empathetic conversational support, crisis detection, and location-aware therapist recommendations.
- Implemented a tool-using LLM agent (LangGraph / REAct) with multi-channel frontends (Streamlit web chat + Twilio WhatsApp) and integrated APIs for real-time actionable outputs.

Publications & Research

Dora, S. et al. (2021). "Detecting Bias in the Relative Grading System Using Machine Learning." *IEEE International Conference on Computing and Communications Technologies.*